

UNIT-2

INTERNAL COMBUSTION ENGINES

Heat engines are otherwise called Thermal Engines. It is a machine which converts heat energy into useful mechanical work. Heat engines develop more than 80% the energy generated in the world.

Heat engines can be broadly classified into two categories

- (i) **External combustion engine:** An engine in which combustion of fuel takes place outside the engine cylinder is called external combustion engine. These engines are generally called EC engines.

Ex: Steam engines, steam turbines, closed cycle gas turbine etc.

- (ii) **Internal combustion engine:** An engine in which combustion of fuel takes place inside the engine cylinder is called internal combustion engine. These engines are generally called IC engines.

Ex: Petrol engine, diesel engine, gas engine etc.

CLASSIFICATION OF IC ENGINES

Internal combustion engines are classified according to

1) According to thermodynamic cycle

- i) Otto cycle ii) Diesel cycle iii) Dual combustion cycle

2) According to number of strokes

- i) Two stroke ii) Four stroke

3) According to number of cylinders

- i) Single cylinder engine ii) Multi cylinder engine

4) According to method of ignition

- i) Spark ignition (petrol, gas engine)
ii) Compression ignition (diesel, vegetable oil)

5) According to type of fuel used

- i) Petrol ii) Diesel iii) Gas
iv) Bio fuel (peanut oil, sunflower oil, linseed oil)

6) According to position of cylinder

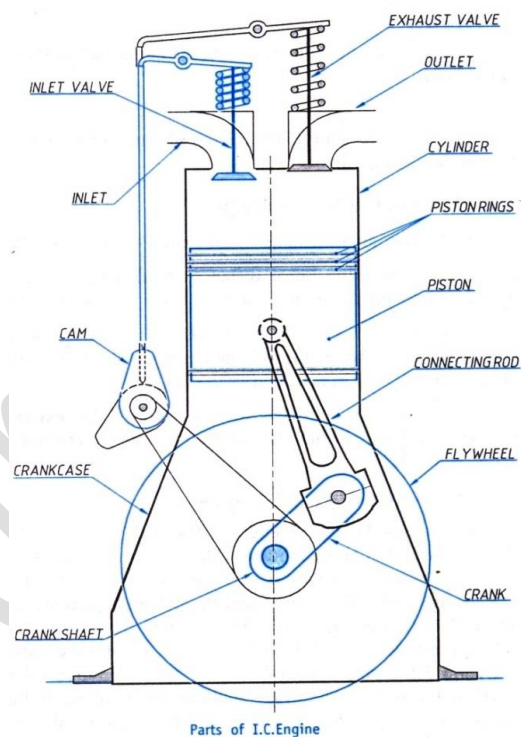
- i) Horizontal engine (hero Honda)
ii) Vertical engine (car, bus, truck engines)

- iii) Vee engine
- iv) Radial engine (old aeroplane engine)
- v) Opposed cylinder engine
- 7) According to method of cooling
 - i) Air cooling ii) Water cooling iii) Liquid cooling
- 8) According to speed of engine
 - i) Slow speed engine ii) Medium speed engine iii) High speed engine

PARTS OF INTERNAL COMBUSTION ENGINE

The parts of internal combustion engines are Cylinder, head, piston, piston rings (compression rings and oil control ring), valves, connecting rod and crankshaft.

Function of IC engine parts:



Cylinder: (cylinder block) The cylinder is the main part of an engine. The combustion takes place in the combustion chamber and these gases exert pressure on the piston, due to high gas pressures the piston reciprocates in the cylinder block. The cylinder is designed to withstand high gas pressure. The temperature in the combustion chamber (cylinder block) will reach up to 2800°C . The cylinder has to be cooled properly either by air cooling or water cooling. In case of air cooled engines fins are provided around the cylinder block (Scooter and bikes) in

water cooled engines water jackets are provided for the circulation of water to carry away the heat around the cylinder block. The cylinder block material; is aluminium alloy

Head: (cylinder head) The head is fitted on the top of the cylinder block. In two stroke engines only spark plug is fitted in the cylinder head.

In four stroke engine inlet valve , exhaust valve and spark is fitted in the cylinder head.- The head is built with two ports, one port, which allows the charge in to the cylinder block, is known as inlet port and the second port that allows the exhaust gases to leave the cylinder block is known as exhaust port. In case of petrol engine spark plug is fitted in the head, in diesel engines fuel injector is fitted to inject the diesel into the cylinder block. The cylinder head material; is aluminium alloy

Piston: The piston is a cylindrical plug, which converts heat energy in to mechanical energy. A two stroke piston is fitted with only compression ring. In four-stroke engine both compression ring and oil control rings are fitted. The piston is connecting to the small end of the connecting rod. The piston is made of aluminium alloy. Functions of piston are

- i) The piston will act as a seal
- ii) To provide the passage for heat flow from piston to cylinder block through rings.
- iii) It transmits the force of explosion to the crankshaft through connecting rod.

Piston rings: The piston rings are the metallic rings inserted into the circumferential grooves provided at the top end of the piston. These rings maintain a gas-tight joint between the piston and the cylinder while the piston is reciprocating in the cylinder.

Valves: the valves are the device which controls the flow of the intake and the exhaust gases to and from the engine cylinder. They are also called poppet valves. These valves are operated by means of cams driven by the crankshaft through timing gear or chain.

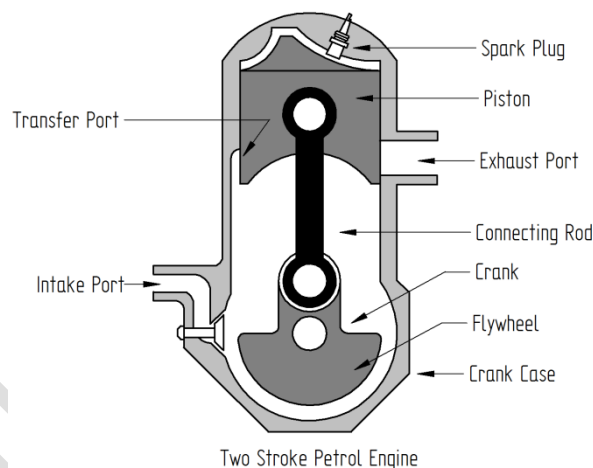
Connecting rod: The small end of the connecting rod is connected to the piston and the big end of the connecting rod is connecting to the crankshaft. The connecting rod converts the reciprocating motion of piston in to rotary motion of crankshaft. The connecting rod is made of I-beam cross section to provide maximum rigidity with minimum weight.

Crankshaft: The big end of the connecting rod is connected to the crankshaft. The power transmission starts from the crankshaft. The crankshaft is rigidly fixed in the crankcase. The other end of the crankshaft is connected to a clutch.

Crankcase: Crankcase is fitted at the bottom of the cylinder block. Two-stroke engine crankcase is properly sealed and made airtight. Four stroke engine crankcase will serve as a reservoir, filled with sufficient quantity of lubricating oil. This oil lubricates the main bearings of crankshaft, big end bearings of connecting rod, lubricates the cylinder liner, piston and piston rings.

WORKING OF TWO STROKE PETROL ENGINE.

The two-stroke petrol engine works on the principle of Otto cycle. The parts of two-stroke petrol engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, spark plug, inlet port, transfer port and exhaust port.



The piston performs two strokes to complete one cycle. The two strokes are

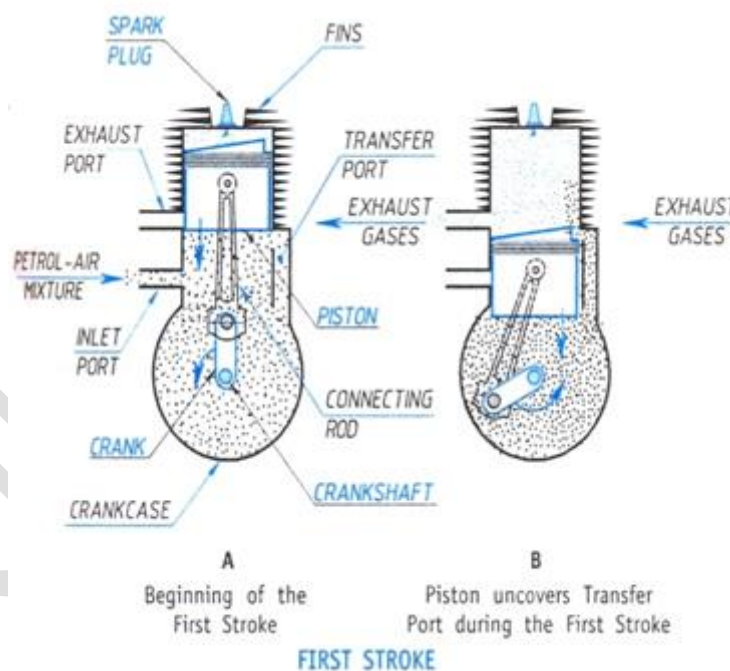
- i) First stroke or down ward
- ii) Second stroke or upward stroke.

First stroke or Down ward stroke or exhaust stroke: At the beginning of this stroke, the piston is in the TDC as shown in the figure (a). At this position, inlet port is opened and hence fresh air petrol mixture enters into the crank case. At this position, compressed air-petrol mixture present in the cylinder in the previous cycle is ignited by the spark generated by the spark plug. The combustion of fuel releases hot gases which increases the pressure in

the cylinder. The high pressure gases exert a pressure on the piston and hence the piston moves from TDC to BDC. Thus piston performs power stroke. The power impulse is transmitted from the piston to the crankshaft through the connecting rod. This causes the crankshaft to rotate at high speeds. Thus work is obtained in this stroke.

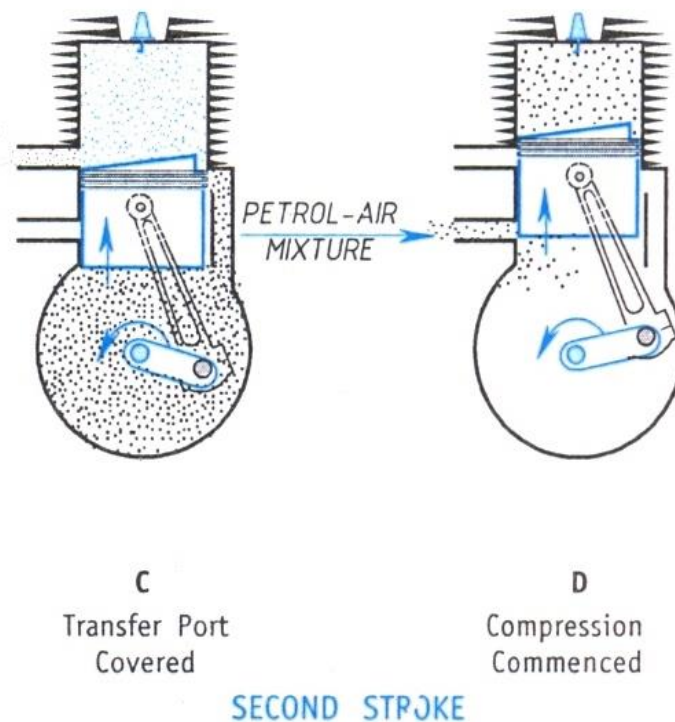
As the piston moves downwards, it uncovers the exhaust port and hence burnt gases escape out of the cylinder as shown in the figure (b). As piston moves downwards further, opens the transfer port and the charge in the crank case is compressed by the underside of the piston as shown in figure. (b). The compressed charge from the crankcase rushes into the cylinder through the transfer port as shown in fig. (c). The charge entering the cylinder drives away the remaining exhaust gases through the exhaust port.

The process of removing the exhaust gases with the help of fresh charge is known as scavenging. The piston is provided with a projection at its top known as 'deflector'. The purpose of providing a deflector is to deflect the fresh charge coming through the transfer port to move towards the top end of the cylinder. By doing this, the fresh charge will be able to drive the entire burnt gases out of the cylinder.



Upward stroke or working stroke: In two-stroke petrol engine some charge is present either in the cylinder block or in the crankcase. To start a two-stroke engine, power is supplied either by using a kicker or by electric start.

During upward stroke, the piston reciprocates from top dead center to bottom dead center. As the piston moves upward volume below the piston increases results in decrease in pressure in the crankcase. Due to pressure difference charge (petrol & air) is drawn from the carburettor. As the piston moves further upwards covers both exhaust and transfer port, now the charge is subjected to compression. Before the end of the compression stroke the spark (crank angle 20° before TDC) occurs in the combustion chamber. Due to combustion of charge, the pressure increase, which pushes the piston downwards i.e. the working stroke of the piston. As the piston moves rapidly in the down ward direction compresses the charge present in the crankcase



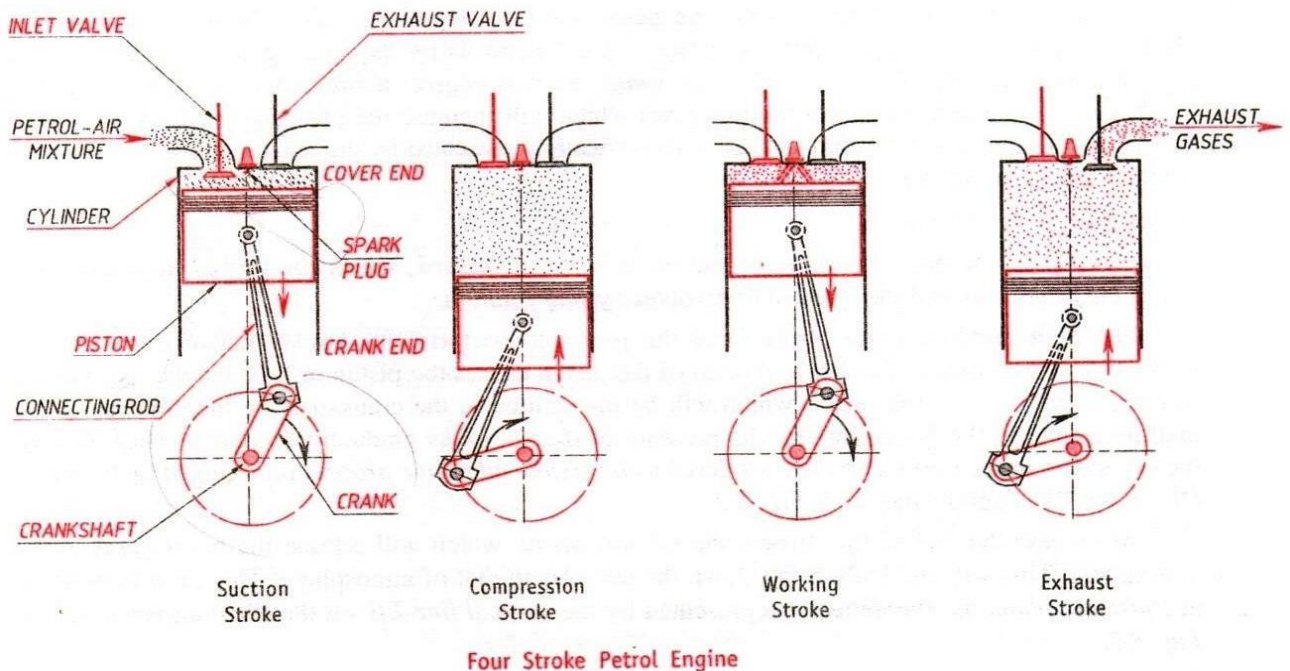
WORKING OF FOUR STROKE PETROL ENGINE

The four-stroke petrol engine works on the principle of Otto (constant volume) cycle. The parts of four-stroke petrol engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, spark plug, inlet and exhaust valve. The four-stroke petrol engine may be air-cooled or water-cooled. The piston performs four strokes to complete one cycle.

The four different strokes are

- 1) Suction stroke 2) Compression stroke 3) Power or Expansion stroke

4) Exhaust stroke.



1. Suction stroke: The suction stroke is completed by rotating the crankshaft from 0° to 180° . During suction stroke the inlet valve opens and exhaust valve should be kept in closed condition. When the piston starts moving from TDC to BDC, the volume above the piston increases, results in decrease in pressure (vacuum). This decrease in pressure draws the petrol and air mixture from the carburettor and delivered it to the cylinder, this process is continuous till the pressure inside the cylinder becomes equal to atmosphere. At the end of suction stroke the cylinder is completely filled with petrol and air mixture. At the end of suction stroke the inlet valve closes. The line AB in the PV diagram represents suction stroke (volume of mixture filled in the cylinder).

2. Compression stroke: Rotation of crankshaft from 180° to 360° completes the compression stroke. During compression stroke both inlet and exhaust valves are closed. In this stroke the piston travels from BDC to TDC. When the piston starts moving from BDC to TDC the mixture is compressed, and the pressure increases in the cylinder. The line BC represents the compression stroke.

At or near the before the end of the compression stroke, the spark occurs, this spark ignites the petrol and air mix. The combustion of mixture releases hot gases, which will increase

pressure at constant volume. The line CD represents increase in the pressure at constant volume.

3.Power stroke: Rotation of crankshaft from 360^0 to 540^0 completes the power stroke.

During power stroke (expansion stroke) both inlet valve and exhaust valve are in closed position. The high-pressure gases produced due to combustion, will exert pressure on the top face of the piston, the piston moves rapidly in the down ward direction performs power stroke.

4.Exhaust stroke: Rotation of crankshaft from 540^0 to 720^0 completes the exhaust stroke. At the beginning of exhaust stroke, the exhaust valve opens, and the upward movement of the piston pushes the exhaust gases out the cylinder. At the end of the exhaust stroke the exhaust valve closes.

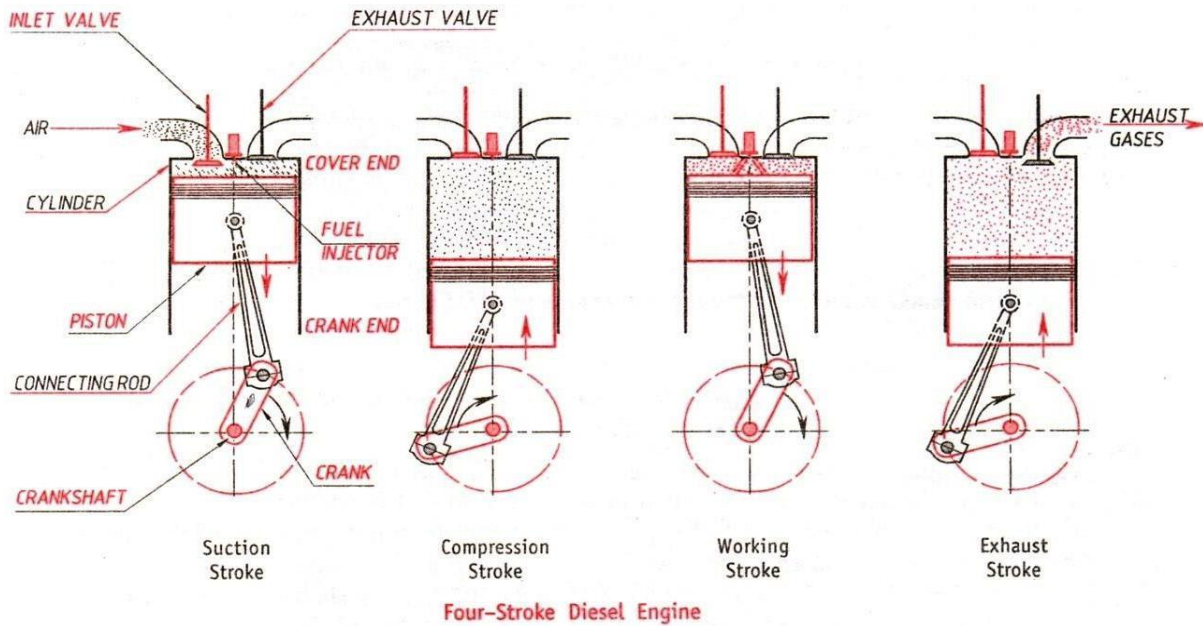
Thus one cycle completes by rotating the crankshaft from 00 to 7200.

WORKING OF FOUR STROKE DIESEL ENGINE

The four-stroke diesel engine works on the principle of diesel (constant pressure) cycle. The parts of four-stroke diesel engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, fuel injector, inlet and exhaust valve. The four-stroke diesel engine may be air-cooled or water-cooled. The piston performs four strokes to complete one cycle.

The four different strokes are

- 1) Suction stroke
- 2) Compression stroke
- 3) Power or Expansion stroke
- 4) Exhaust stroke.



1) **Suction stroke:** The suction stroke is completed by rotating the crankshaft from 0° to 180° . During suction stroke the inlet valve opens and exhaust valve should be kept in closed position. When the piston starts moving from TDC to BDC, the volume above the piston increases, resulting in a decrease in pressure (vacuum). This decrease in pressure draws the air from the atmosphere and fills the air into the cylinder; this process is continuous till the pressure inside the cylinder becomes equal to atmosphere. At the end of suction stroke the cylinder is completely filled with air. At the end of suction stroke the inlet valve closes. The line AB in the PV diagram represents suction stroke (Volume of air filled in the cylinder).

2) **Compression stroke:** Rotation of crankshaft from 180° to 360° completes the compression stroke. During compression stroke both inlet and exhaust valves are closed. In this stroke the piston travels from BDC to TDC. When the piston starts moving from BDC to TDC, the air is compressed, both pressure and temperature of the air increase. At the end of the compression stroke the temperature of the air reaches the ignition temperature of diesel. The line BC represents the compression stroke.

Before the end of the compression stroke, the fuel injector starts injecting the diesel into the combustion chamber. The heat of the compressed air burns the injected diesel. The combustion takes place at constant pressure. The line CD represents increase in the pressure at constant pressure.

3) **Power stroke:** Rotation of crankshaft from 360° to 540° completes the power stroke.

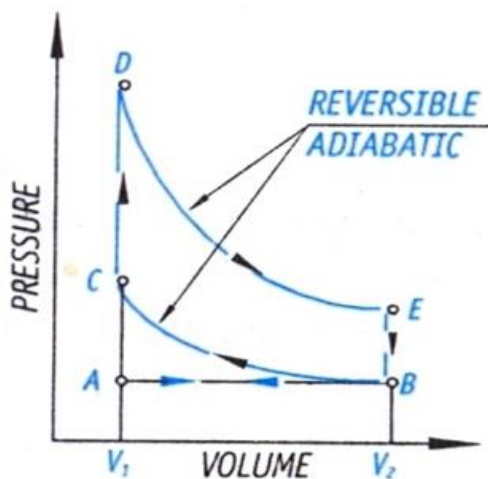
During power stroke (expansion stroke) both inlet valve and exhaust valve are in closed position. The high-pressure gases produced during combustion, will exert pressure on the top face of the piston, the piston moves rapidly in the down ward direction performs power stroke. The energy is supplied to the flywheel during power stroke. This energy propels the vehicle

4) **Exhaust stroke:** Rotation of crankshaft from 540° to 720° completes the exhaust stroke. At the beginning of exhaust stroke, the exhaust valve opens, and the upward movement of the piston pushes the exhaust gases out the cylinder. At the end of the exhaust stroke the exhaust valve closes.

Thus one cycle completes by rotating the crankshaft from 0° to 720° , i.e., two revolutions of the crankshaft.

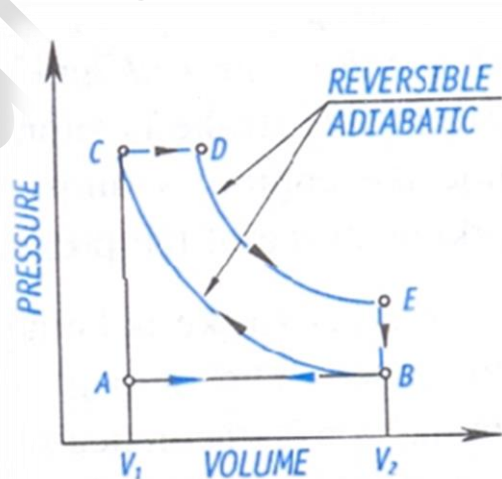
P-V DIAGRAMS

OTTO CYCLE



Theoretical Otto Cycle

DIESEL CYCLE



Theoretical Diesel Cycle

COMPARISON BETWEEN PETROL ENGINE AND DIESEL ENGINES.

SI. No	Petrol Engine (SI Engine)	Diesel Engine (CI Engine)
1.	Draws a mixture of petrol and air during suction stroke .	Draws only air during suction stroke.
2.	The carburetor is employed to mix air and petrol in the required proportion and to supply it to the engine during suction stroke.	The injector is employed to inject the fuel at the end of compression stroke.
3.	Compression ratio ranges from 7: 1 to 12: 1	Compression ratio ranges from 18:1 to 22:1
4.	The charge (Le petrol and air mixture) is ignited with the help of spark plug. This type of ignition is called spark	The ignition of the diesel is accomplished by the compressed air which will have been heated due
5.	The combustion of fuel takes place at constant volume.	The combustion of fuel takes place at constant pressure.
6.	Works on theoretical Otto Cycle.	Works on theoretical Diesel Cycle.
7.	Power developed is less.	Power developed is more.
8.	Thermal efficiency is low. It is up to about 26%	Thermal efficiency is high. It is up to about 40%.
9.	These are high speed engines	These are low speed engines.
10.	The maintenance cost is less.	The maintenance cost is more.
11.	The running cost is high because of the higher cost of petrol.	The running cost is low because of lower cost of diesel
12.	Lighter and cheaper because of low compression ratio	Heavier and costlier because of high Compression ratio.

COMPARISON BETWEEN 2-STROKE AND 4-STROKE I.C. ENGINES.

SI. No	2-Stroke Engine	4-Stroke Engine
1.	Requires two separate strokes to complete one cycle of operation.	Requires four separate strokes to complete one cycle of operation.
2.	Power is developed in every revolution of the crankshaft	Power is developed for every revolutions of the crankshaft.
3.	The inlet, transfer and exhaust ports are opened and closed by the movement of piston itself.	The inlet and exhaust are opened and closed by the valves.
4.	Tuning moment is not uniform and hence requires a heavier flywheel.	Tuning moment is uniform and hence Requires lighter flywheel.
5.	The charge is first admitted into the crankcase and then transferred to the Engine cylinder.	The charge is directly admitted in to the engine cylinder during the suction Stroke.
6.	For the same power developed the Engine is heavy and bulky.	For the same power developed the Engine is light and compact.
7.	Thermal efficiency is low.	Thermal efficiency is high.
8.	Requires greater lubricant and coolant.	Requires lesser lubricant and coolant.
9.	Fuel consumption is more.	Fuel consumption is less.
10.	Initial cost is less.	Initial cost is more.

Simple calculations in internal combustion engines.

i) **Mean effective pressure (p_m):** It is defined as the average pressure is acting on the piston during the entire expansion (power stroke) stroke.

$$P_m = \frac{s a}{l}$$

Where,

p_m = Mean effective pressure N / m²

s = spring constant of the spring used in the piston indicator,

l = length of the indicator diagram,

a = area of the indicator diagram.

Note that, spring constant is the pressure required to cause unit deflection of the spring.

ii) **Indicated power (IP):** The power developed within the piston –cylinder arrangement by the combustion of fuel is known as the indicated power.

i) When p_m is expressed in N/ m²

$$IP = \frac{P_m L A n}{60 * 1000} \quad \text{kW}$$

ii) When p_m is expressed in **bar**

$$IP = \frac{100 P_m L A n}{60} \quad \text{kW}$$

Where,

P_m = mean effective pressure.

L = stroke length.

A = area of cross-section of the piston.

n = number of cycles per minute.

= N/2 for a four stroke engine.

= N for a two stroke engine.

N = crank shaft speed, rpm

ii) **Brake power (BP):** The power available at the crank shaft is always less than the power developed within the piston-cylinder arrangement because of frictional losses in the moving parts. The power actually available at the crank shaft is called the brake power. It can be measured using *dynamometers*. One such dynamometer is the *brake-drum* dynamometer.

The torque on the brake drum is given by,

$$T = (W - S) * R \quad \text{Nm}$$

Where,

W = weight on the rope, N.

S = spring balance reading, N.

R = mean radius of brake drum, m.

Brake power is given by.

$$BP = \frac{2\pi NT}{60 * 1000} \quad \text{kW}$$

iii) **Frictional power:** The difference between indicated power and brake power is known as frictional power

$$FP = (IP - BP) \quad \text{kW}$$

iv) **Mechanical efficiency:** It is defined as the ratio of brake power to indicated power

$$\eta_{\text{mech}} = \frac{BP}{IP}$$

v) **Thermal efficiency:** In IC engines, energy is supplied to the engine by burning fuel. But all of the energy that is supplied is not converted into useful mechanical work. The thermal efficiency can be calculated either for the indicated power, or for the brake power. Accordingly they are referred to as indicated thermal efficiency, and brake thermal efficiency.

Heat supplied to the engine per sec = mass of fuel burnt (m_f) x calorific value (CV).

1. Indicated thermal efficiency

$$\eta_{\text{indicated thermal}} = \frac{\text{Indicated power}}{m_f * CV}$$

2. Brake thermal efficiency

$$\eta_{\text{brake thermal}} = \frac{\text{Brake power}}{m_f * CV}$$

Where,

m_f = Mass of fuel used in **kg / sec.**

CV = Calorific value of fuel **kJ / kg**

3. Specific fuel consumption (SFC):

It is the mass of fuel supplied per hour in order to get unit power output.

$$\text{SFC} = \frac{m_f}{\text{Power}} \frac{\text{kg}}{\text{kW}} - \text{hr}$$

SFC can be calculated on indicated power basis or on brake power basis.

***Problems are discussed in the classes**

❖ Introduction to Electric vehicles

An electric vehicle (EV) is a vehicle powered by an electric motor, instead of an internal combustion engine (ICE), and the motor is run using the power stored in the batteries. The batteries have to be charged frequently by plugging into any main (120 V or 240 V) supply. EV has a much longer history than most people realize. EVs were seen soon after Joseph Henry introduced the first DC-powered motor in 1830. The first known electric car was a small model built by Professor Stratingh in the Dutch town of Groningen in 1835. The first EV was built by in 1834 by Thomas Davenport in the U.S., followed by Moses Farmer, who built the first two-passenger EV in 1847. There were no rechargeable electric cells (batteries) at that time. An EV did not become a viable option until the Frenchmen Gaston Plante and Camille Faure respectively invented (1865) and improved (1881) the storage battery.

EVs are known as zero emissions vehicles (ZEVs) and are much environment friendly than gasoline- or LPG-powered vehicles. As EVs have fewer moving parts, maintenance is also minimal. With no engine there are no oil changes, tune-ups, or timing and there is no exhaust. EVs are also far more energy efficient than gasoline engines and they are very quiet in operation.

EVs have been in continual use since the 1900s in various applications. Today these quiet vehicles with no tailpipe emissions are no longer limited to golf carts. New advances in battery technology, system integration, aerodynamics, and research and development by major vehicle manufacturers has led to the producing of electric vehicles that will play a practical role on city streets.

❖ Engineering philosophy of EVs

Engineering integrates science, technology and management to solve real world problems effectively and economically, while usually asks for 'how' and 'what for', pursues 'best solution' and emphasizes 'teamwork'. As EV engineering is multidisciplinary, specialists in those areas of engineering must work together; electrical engineers, electronic engineers, mechanical engineers, chemical engineers and mainstream automotive engineers must pool

their knowledge in the main areas that must be integrated: body design, electric propulsion, energy sources, energy refuelling and energy management.

❖ EV CONCEPT

The modern EV concept is summarized as follows:

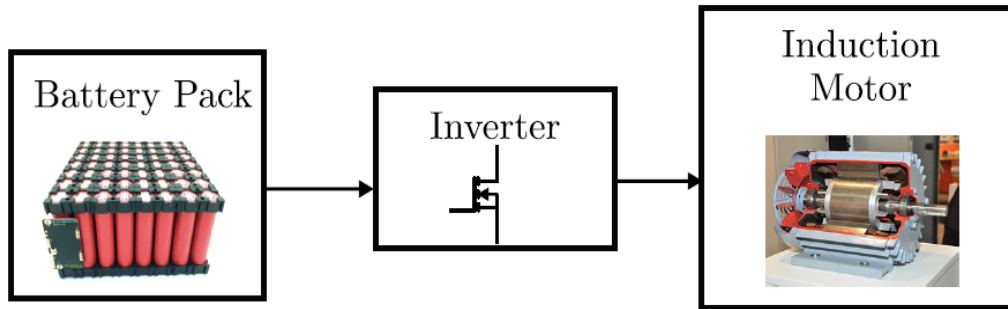
- The EV is a road vehicle based on modern electric propulsion which consists of the electric motor, power converter and energy source, and it has its own distinct characteristics;
- The EV is not just a car but a new system for our society, realizing clean and efficient road transportation;
- The EV system is an intelligent system which can readily be integrated with modern transportation networks;
- EV design involves the integration of art and engineering;
- EV operating conditions and duty cycles must be newly defined;
- EV users' expectations must be studied, hence appropriate education must be conducted.

❖ What are Electric Vehicles (EVs)?

Electric vehicles use an electric motor for propulsion rather than using an internal combustion engine. Electric vehicles have a battery that can be charged by electric supply. This electric energy is used to run the motor. Electric vehicles can be fully electric or hybrid type, which means a combination of electric motor and combustion engine.

An Electric Vehicle contains 3 main parts

(1) Energy Source. (2) Power Converter. (3) Traction Motor.



Battery Pack-Energy source

The most common battery type in modern electric vehicles are lithium-ion and lithium polymer, because of their high energy density compared to their weight. Other types of rechargeable batteries used in electric vehicles include lead-acid ("flooded", deep-cycle, and valve regulated lead acid), nickel-cadmium, nickel-metal hydride, and, less commonly, zinc-air, and sodium nickel chloride ("zebra") batteries.[1] The amount of electricity (i.e. electric charge) stored in batteries is measured in ampere hours or in coulombs, with the total energy often measured in kilowatt-hours.

Inverter

An inverter is a device that converts DC power to the AC power used in an electric vehicle motor. The inverter can change the speed at which the motor rotates by adjusting the frequency of the alternating current. It can also increase or decrease the power or torque of the motor by adjusting the amplitude of the signal.

Motor

Using power from the traction battery pack, this motor drives the vehicle's wheels. Some vehicles use motor generators that perform both the drive and regeneration functions.

Both AC and DC motors use electrical current to produce rotating magnetic fields that, in turn, generate rotational mechanical force in the armature—located on the rotor or stator—around the shaft. The various motor designs use this same basic concept to convert electric energy into powerful bursts of force and provide dynamic levels of speed or power.

Main Motor Components

While electric motors may differ from one design or type to another, many of them contain these parts and assemblies (located from the center working outward):

- Central motor shaft
- Windings
- Bearings (to reduce friction and wear)

- Armature (located on the rotor, the rotating part, or the stator, the stationary part)
- Brushes (in DC motors)
- Terminals
- Frame and end shields

❖ **Need of Electric Vehicles (EVs) Or Importance of Electric vehicle**

- Carbon monoxide, sulphur dioxide, hydrocarbons are some of the gases which cause air pollution. These gases are the result of emission from an internal combustion engine. According to a report by WHO, India is home for 14 out of 20 most polluted cities in the world.
- All these pollution-causing gases and emission is harmful to people like women, children and elderly people. This emission will also lead to global warming.
- On the other side electric vehicles are environmentally friendly and do not emit toxic gases, it will help reduce the global warming effect.
- It will help to fulfil the commitment of the Paris climate agreement.
- It will be helpful in reducing oil imports. India is 5th largest importer of oil with consumption of 2.2 million barrels a day.
- Electric vehicles are cost-effective in maintenance (because of less fluid and less moving parts). Government is also providing subsidies on vehicles and lower taxes on an electric motor.

❖ **Some drawbacks of Electric Vehicles**

- Electric engines provide limited mileage. A complete recharge will take a lot of time and fewer infrastructures for charging across the country.
- It will be poor for highway driving; the top speed of the electric car is 70mph.
- Due to the high cost of lithium-ion battery and the use of electricity will make the overall cost of the electric vehicle high.
- A small amount of pollution is also caused by electric vehicles because of the toxicity of batteries and all electricity origin is also not renewable.

❖ Challenges related to the EV industry in India

- Electric vehicles are not penetrated in the Indian market, almost the lowest in the world.
- India is having challenges related to technological production of electronics such as the battery, semiconductors etc.
- Government's uncertainty in policymaking is another challenge for investors in this industry. It is a capital-intensive sector and will make a profit in the near future.
- The local production is just 35% of the total input production of electric vehicles.
- India does not have any reserve for lithium and cobalt which is important for battery production which is the most important component for EVs. This is related to increasing the cost of production because of dependency on China and Japan.
- High rate of GST and depreciation of rupee is another challenge for the EV industry.
- Lack of infrastructure related to AC versus DC charging station, grid stability, fear anxiety-related to the battery will run out soon, are other hindrances.
- This industry needs a skilled worker and high services.
- India lacks dedicated courses and training in this area.

Regenerative Braking System For Automobile

Every time we step on our car's brakes, we are actually wasting energy. As we know that energy can neither be created nor be destroyed. It can be just converted from one form to another. So when our car slows down, the kinetic energy that was propelling it in the forward direction has to go somewhere. Most of it simply gets released in the form of heat and becomes useless. That energy, which could have been used to do work, is essentially wasted. The solution for this kind of this problem is Regenerative Braking System.

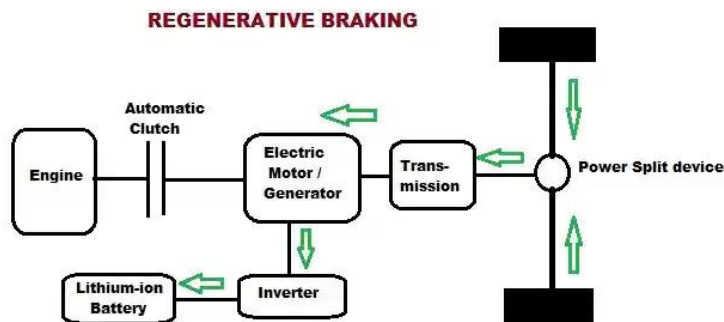
This is a new type of braking system that can recollect much of the car's kinetic energy and convert it into electrical energy or mechanical energy. Regenerative braking is one of the emerging technologies of automotive industry which can prove to be very beneficent. Using regenerative braking system in a vehicle not only results in the recovery of the energy but it also increases the efficiency of vehicle (in case of hybrid vehicles) and saves energy, which is stored in the auxiliary battery.

WORKING PRINCIPLE:

We have to fabricate this model, when you are applied the brake condition while some energy is going loss that is conditions to avoiding, we have to fabricate on kinetic energy storage method, which is contain for like sun and planetary gear and scroll spring also attached.

When you need the brake, your applied the brake and both of clutch also applied. Brake condition done stop the wheel and disc contain system and clutch condition is disengaged the wheel and axle that is condition energy going loss, whereas our model engage the ring gear is rotate the reverse direction, as well as connecting on scroll spring rotate also same direction that is condition save the kinetic energy on clutch applied condition.

When you release the clutch, regenerating the kinetic energy on axle which is condition increased the pickup and fuel efficiency.



ADVANTAGE

- Improved Performance.
- Improved Fuel Economy - Dependent on duty cycles, power train design, control strategy, and the efficiency of individual components.
- Reduction in Engine wears.
- Reduction in Brake Wear- Reducing cost of replacement brake linings, cost of labor to install them, and vehicle down time,
- Emissions reduction- engine emissions reduced by engine decoupling, reducing total engine revolutions and total time of engine operation.
- Operating range is comparable with conventional vehicles- a problem not yet overcome by electric vehicles.

APPLICATION:

- For recovering Kinetic energy of vehicle lost during braking process.
- One theoretical application of regenerative braking would be in a manufacturing plant that moves material from one workstation to another on a conveyer system that stops at each point.
- Regenerative braking is used in some elevator and crane hoist motors.

- Regenerative Braking Systems are also used in electric railway vehicle (London Underground & Virgin Trains).

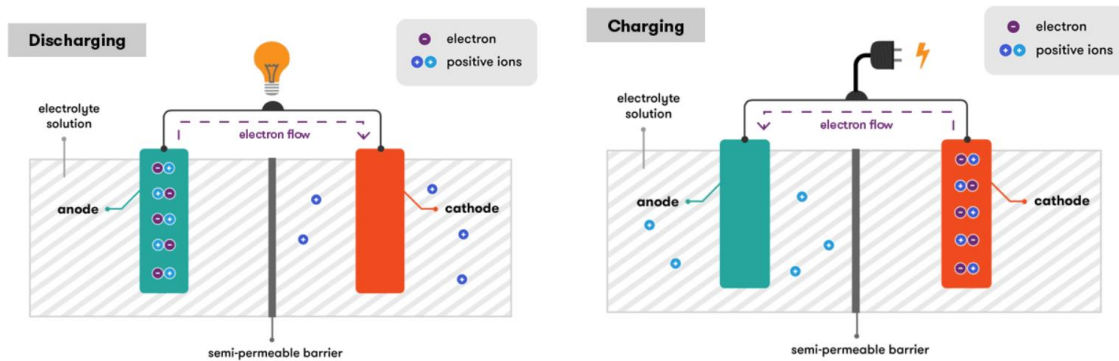
BATTERY

TYPES OF BATTERIES

- **Deep-Cycle Lead-Acid**
- **Horizon Lead-Acid**
- **Nickel Cadmium/Nickel Iron**
- **Nickel-Metal Hydride**
- **Sodium Sulphur-Sodium sulfur (NaS)**
- **Sodium Nickel Chloride**
- **Lithium-Ion**
- **Lithium Metal Sulphide**
- **Lithium-Polymer**

DEEP-CYCLE LEAD-ACID

- The Pb acid battery has the nominal cell voltage of 2V, specific energy of 35Wh/kg, energy density of 70Wh/l, and specific density of 200W/kg. It uses metallic lead as the negative electrode and lead dioxide as the positive electrode.
- During discharge, the negative electrode performs oxidation reaction which drives electrons to the external circuit, while the positive electrode carries out reduction reaction which accepts electrons from the external circuit.
- Electrolyte is Distilled water + H_2SO_4
- The electrons flow from one electrode called the anode (or negative electrode) to another electrode called the cathode (the positive electrode).
- The technical chemical term for a reaction that involves the exchange of electrons is a reduction-oxidation reaction, more commonly called a redox reaction
- Reduction is the gain of electrons, and is what occurs at the cathode; we say that the cathode is reduced during the reaction.
- Oxidation is the loss of electrons, so we say that the anode is oxidised.



- During charge, the process is reversed so that electrons are injected into the negative electrode to perform reduction while the positive electrode releases electrons to carry out oxidation.

Advantages of deep-cycle Lead-acid Batteries:

- Proven technology and mature manufacturing,
- Low cost (lowest of all secondary batteries),
- High cell voltage (highest of all aqueous electrolyte batteries),
- Good high-rate performance that suitable for EV applications,
- Good low- and high-temperature performances,
- High energy efficiency (75-80%), and
- Available in a variety of sizes and designs.

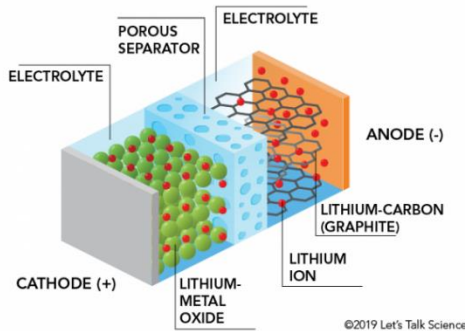
Disadvantages of Deep-cycle Lead-acid Batteries:

- Its specific energy and energy density are relatively low (typically 35 Wh/kg and 70 Wh/l)
- Its self-discharge rate is relatively high (about 1 % per day at 25 degree Celsius),
- Its cycle life is relatively short (about 500 cycles), and
- It is unsuitable for long-term storage because of electrode corrosion by sulphation.

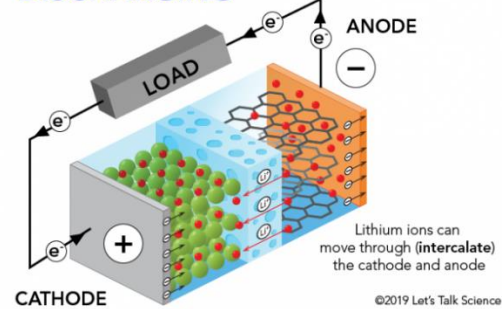
2. Lithium-Ion

- The lithium-ion battery was predicted to be a long-term battery. The battery is being developed for Nissan by Sony Corporation.
- The battery is 28.8 V and consists of eight metal cylindrical cells encased in a resin module. Each battery has a built-in cell controller to ensure that each cell is operating within a specified voltage range of 2.5 to 4.2 V during charging and discharging

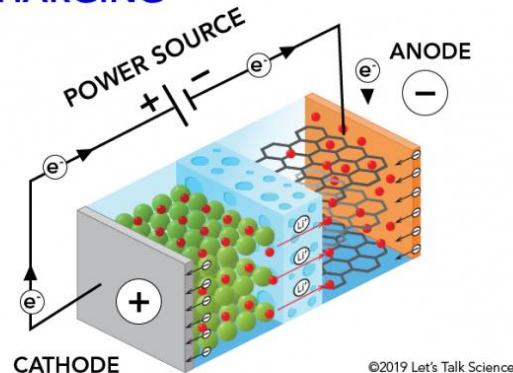
PARTS OF A LITHIUM-ION BATTERY



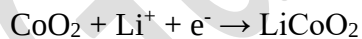
DISCHARGING



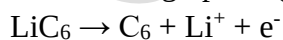
CHARGING



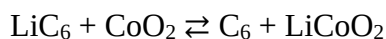
Reduction takes place at the cathode. There, cobalt oxide combines with lithium ions to form lithium-cobalt oxide (LiCoO_2). The half-reaction is:



Oxidation takes place at the anode. There, the graphite intercalation compound LiC_6 forms graphite (C_6) and lithium ions. The half-reaction is:



Here is the full reaction (left to right = discharging, right to left = charging):



Advantages

- It's simple. lithium-ion batteries have the highest charge density of any comparable system. This means they can give you a ton of energy without being very heavy.
- Lithium is also the lightest of all metals
- This is for two reasons. First, lithium is the most electropositive element. Electropositivity is a measure of how easily an element can donate electrons to produce

positive ions. In other words, it's a measure of how easily an element can produce energy. Lithium loses electrons very easily. This means it can easily produce a lot of energy.

- High specific energy of 100 Wh/kg,
- High specific power of 300 W/kg,
- Battery life of 1000 cycles.

Disadvantages

- Very high cost and the ventilation system required to keep the batteries cool.
- The manufacturing costs are high because the battery uses an oxidized cobalt material for the anode, a highly purified organic material for the electrolyte, and a complex cell control system.

3. Nickel-Metal Hydride Battery

A nickel metal hydride battery, abbreviated NiMH or Ni-MH, is a type of rechargeable battery.

The chemical reaction at the positive electrode is similar to that of the nickel-cadmium cell (NiCd), with both using nickel oxide hydroxide (NiOOH). However, the negative electrodes use a hydrogen-absorbing alloy instead of cadmium. A NiMH battery can have two to three times the capacity of an equivalent size NiCd, and its energy density can approach that of a lithium-ion battery.

A nickel-metal hydride battery is composed of nontoxic recyclable materials and is environmentally friendly.

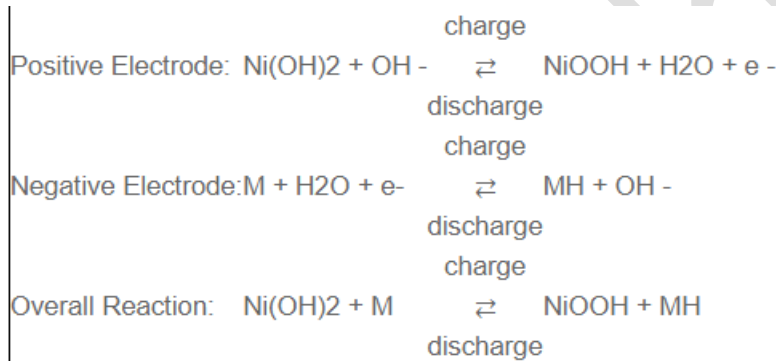
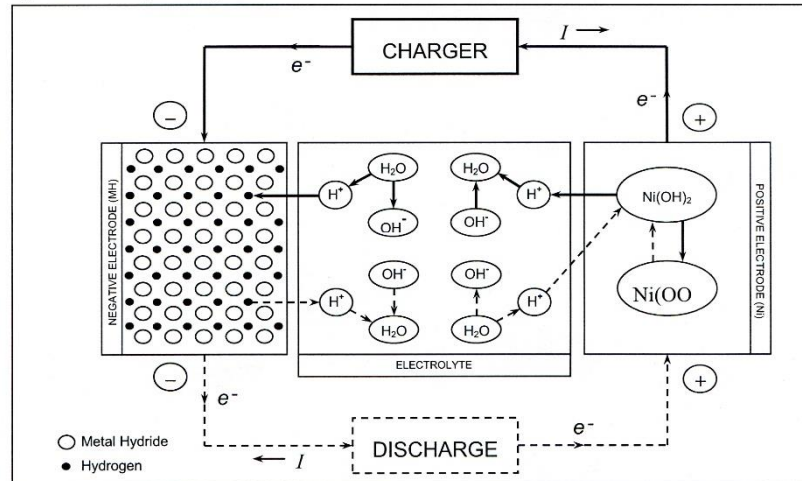
The NiMH has twice the range and cycle life of today's lead-acid batteries.

It is composed of nickel hydroxide and a multicomponent, engineered alloy consisting of vanadium, titanium, nickel, and other metals.

It is sealed, is maintenance free, and can be charged as quickly as 15 minutes. It can withstand overcharging and over discharge abuse.

Ovonic Battery of Troy, MI, is currently manufacturing the NiMH battery.

Figure 3.3 Transport Diagram



Charging

When a NiMH cell is charged, the positive electrode releases hydrogen into the electrolyte. The hydrogen in turn is absorbed and stored in the negative electrode. The reaction begins when the nickel hydroxide (Ni(OH)_2) in the positive electrode and hydroxide (OH^-) from the electrolyte combine. This produces nickel oxyhydroxide (NiOOH) within the positive electrode, water (H_2O) in the electrolyte, and one free electron (e^-). At the negative electrode the metal alloy (M) in the negative electrode, water (H_2O) from the electrolyte, and an electron (e^-) react to produce metal hydride (MH) in the negative electrode and hydroxide (OH^-) in the electrolyte.

Discharge

When a NiMH cell is discharged, the chemical reactions are the reverse of what occurs when charged. Hydrogen stored in the metal alloy of the negative electrode is released into the electrolyte to form water. This water then releases a hydrogen ion that is absorbed into the positive electrode to form nickel hydroxide